

Fake News Detection Using Machine Learning

Abstract

The rapid growth of social media and online news platforms has significantly increased the spread of fake news, creating serious social, political, and economic challenges. Fake news, defined as false or misleading information presented as legitimate news, can influence public opinion and disrupt societal stability. Detecting such misinformation manually is time-consuming and inefficient, particularly in large-scale digital environments.

This project presents an automated fake news detection system using web scraping and deep learning techniques. Web scraping is employed to collect real-time news articles from trusted online sources, enabling the creation of a dynamic and up-to-date dataset. The system is developed following the Agile methodology to support iterative development, continuous testing, and flexible model improvement.

Natural Language Processing (NLP) techniques, including tokenization, stopword removal, and stemming, are applied to preprocess the collected textual data. The processed text is converted into numerical sequences using tokenization and padding methods suitable for deep learning models.

The proposed system utilizes a Long Short-Term Memory (LSTM) network implemented using TensorFlow and Keras for classification. LSTM is selected due to its capability to capture long-term dependencies and contextual relationships within sequential text data, making it highly effective for news classification tasks. Unlike traditional machine learning models that require manual feature engineering, LSTM automatically learns semantic patterns directly from textual input.

The performance of the model is evaluated using accuracy, precision, recall, and F1-score metrics. Experimental results indicate that the LSTM-based approach achieves improved accuracy and reliability in detecting fake news compared to conventional methods.

The developed system provides an efficient, scalable, and intelligent solution for identifying misinformation and can be extended for real-time deployment in web applications and digital news monitoring platforms.